INTRODUCTION

This consolidated project report presents six interconnected projects demonstrating the development and practical applications of Generative Artificial Intelligence (Generative AI), Large Language Models (LLMs), and Agentic AI. The work follows a progressive engineering approach, beginning with the fundamental mathematical architecture of language models and extending towards the development of autonomous, intelligent applications capable of assisting users, enforcing statutory compliance, and automating multifaceted real-world tasks.

The first project focuses on implementing a decoder-style Transformer language model from scratch using PyTorch. It explores the fundamental building blocks of modern Transformer architecture, including tokenization strategies, learned token embeddings, sinusoidal positional encodings, multi-head self-attention mechanisms, position-wise feed-forward networks, and autoregressive next-token prediction. This implementation provides rigorous practical insight into how neural language models vectorize, attend to, and generate sequence-based representations, establishing a solid foundation for understanding contemporary Generative AI systems.

The second project explores the deployment and operationalization of a locally hosted Large Language Model using the Ollama framework. By integrating a quantized Qwen open-weights model with a custom web-based user interface, the application enables users to interact with an AI model running entirely within an on-premise, local computing environment. This project demonstrates edge deployment, backend REST API integration, and full-stack interface engineering while directly addressing critical enterprise concerns, including data privacy, compute latency, hardware resource constraints, and the strategic reduction of reliance on proprietary cloud-hosted AI providers.

The subsequent four projects extend AI capabilities beyond passive response generation into practical institutional workflow automation through Agentic AI. Unlike conventional chatbots that primarily generate conversational text, goal-oriented AI agents are architected to interpret high-level task instructions, maintain environment state, query external databases, dynamically invoke tools, and execute structured operations within strict organizational boundaries.

The third project involves developing a University Leave Processing Agent to streamline institutional faculty and student leave-management procedures. The system combines natural language intent parsing, institutional business rules, and relational database transactions to assist administrative staff in verifying entitlement balances, evaluating request validity, and logging approved leaves. This application illustrates how conversational agents can simplify repetitive administrative procedures and enhance workflow efficiency.

The fourth project focuses on an AI-powered Recruitment and Placement Preparation Assistant (PlacementPrep). It applies language model reasoning to recruitment-related activities

and student career readiness. The platform provides automated resume critique against job descriptions, simulates domain-specific technical interview questions, assesses candidate conceptual responses, and generates constructive gap analysis to support students in achieving career placement milestones.

The fifth project presents an AI-powered Email Processing Agent that automates inbound enterprise communication workflows through the orchestration of n8n, Google Gmail, and Google Drive. The system categorizes incoming messages, extracts actionable entities, drafts context-aware replies, and files relevant attachments automatically. Crucially, Human-in-the-Loop (HITL) supervisory controls are embedded into the pipeline to ensure that sensitive or outbound communications undergo human verification before execution, maintaining rigorous institutional accountability.

The sixth project culminates in an Intelligent Student Grievance Resolution Agent, an enterprise-grade quasi-judicial portal engineered in strict compliance with the statutory mandates of the UGC (Redressal of Grievances of Students) Regulations, 2023. Rejecting unpredictable generative chatbots in sensitive administrative domains, the system employs an explainable Natural Language Processing triage engine to classify petitions across seven university jurisdictions, dynamically assigns urgency-tiered Service Level Agreements scaling from 24-hour critical turnaround to routine schedules, cross-references statutory bylaws, safeguards petitioner confidentiality through cryptographic identity shielding, and synthesizes authoritative draft resolution decrees for university authorities.

Together, these six interconnected projects span core frontiers of modern computing, including Natural Language Processing, deep learning tensor operations, Transformer mathematics, full-stack web engineering, relational database management, local LLM containerization, workflow orchestration, and regulatory compliance. Each implementation addresses a progressive stage of intelligent application development, establishing a continuous trajectory from neural mechanics to deployed multi-agent enterprise systems.

The report is organized into eleven comprehensive chapters. Chapters 1 and 2 introduce the foundational concepts, explain the project objectives, and present the theoretical technologies utilized. Chapters 3 through 8 describe the six individual implementations in detail, covering system architecture, development methodology, and functional execution. Chapter 9 presents testing protocols, performance observations, and comparative evaluation. Chapter 10 synthesizes overall learning outcomes and engineering proficiencies gained, and Chapter 11 concludes the consolidated work with prospective directions for future research.